

ARC Innovation at Sheba Medical Center: Building the AI and data tools to reimagine healthcare

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

ABOUT ARC



With its data-first
ARC Innovation at
Medical Center is
reimagining global
healthcare; working
Google Cloud and
collaborate with
medical facilities
worldwide.

About ARC

ARC Innovation, founded in 2019 at Sheba Medical Center, one of the world's top hospitals, has developed a structured approach to innovation to equip all parts of the health ecosystem with the tools and resources to advance healthcare delivery and improve outcomes.

Industries: Healthcare, Life Sciences

Google Cloud results

- ARC Innovation at Sheba Medical Center's data-centric approach embraces AI for better decision-making and healthcare delivery, boosting innovation
- Data dashboards show the impact of treatment on ovarian cancer patients
- AI supports and improves critical clinical decisions

Location: Israel

ARC Innovation at Sheba Medical Center collaborates across boundaries

Google Cloud (<https://cloud.google.com/>)

Cloud AutoML (<https://cloud.google.com/automl/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Looker Studio

(<https://cloud.google.com/looker-studio>)

during the
treatment
of ovarian
cancer

The global healthcare sector finds itself under unprecedented pressure post-pandemic. There is a growing crisis in recruitment, with the World Health Organization estimating a shortfall of 18 million healthcare workers

(<https://gh.bmj.com/content/7/6/e009316>) by 2030.

There is also a rapidly aging population around the world, with the number of people 65 or older expected to double from 700 million to 1.5 billion (<https://www.un.org/development/desa/en/news/population/our-world-is-growing-older.html>)

by 2050. The spiraling cost of healthcare and the growing range of complex treatments now available are among the many other factors impacting the sector.

The conventional models of healthcare, which focus primarily on acute medicine, rather than early intervention and prevention, are looking increasingly unscalable and unsustainable when faced with these challenges. If levels of care are to be maintained, let alone improved, the sector needs to open its doors to fresh ideas, wider collaboration, and new data-driven ways of working.

Achieving meaningful and measurable impact

Israel's world-leading ARC Innovation at Sheba Medical Center has placed itself at the forefront of this effort to redefine the sector.

ARC is making a meaningful impact on healthcare by providing clinicians and practitioners with the powerful AI data tools and insights needed to reimagine the industry. The aim is to enable patients to be diagnosed and treated earlier, faster, and more effectively—elevating clinical outcomes, while reducing the level of resource, time, and budget required.

Building a borderless and federated learning model

Unlike conventional medical institutions, which are often burdened by legacy systems and processes, the Sheba Medical Center is globally renowned for its agility and innovation, a trademark of Israel's start-up culture.

ARC is on a journey to build what it calls a "borderless and federated learning model," collaborating with its peers internationally in the [secure training of AI models](https://nephrocage.org/2021/12/federated-learning-enables-better-sharing-of-healthcare-data-for-ai/) (<https://nephrocage.org/2021/12/federated-learning-enables-better-sharing-of-healthcare-data-for-ai/>) and pooling expertise to redefine healthcare in the 21st century.

"ARC stands for accelerate, redesign and collaborate," says Eran Kacen, CTO and COO of ARC. "These are the three fundamental principles that underpin all our work. We can't redesign the future of

medicine on our own. That's why we've established close relationships with healthcare providers all around the world, as well as tech giants and start-ups. Expanding and diversifying our data through sharing is key to success."

The achievements so far

ARC may only be four years old, but its initial achievements are promising. They include:

- A needle-free, hand-held device that tests blood by analyzing tiny blood vessels around the eye. Data is captured in real time, reducing the amount of costly and time-consuming lab work. The device has been used on the [International Space Station](https://www.timesofisrael.com/invented-in-israel-no-needle-blood-test-will-blast-off-for-testing-in-space/) (<https://www.timesofisrael.com/invented-in-israel-no-needle-blood-test-will-blast-off-for-testing-in-space/>)
- .
- An augmented reality solution that directs medical teams during spinal procedures, showing surgeons exactly where to insert a needle. The solution has reduced time in theater from 45 minutes to just 15.
- A military-grade thermal imaging camera, which can "see" through the patient's abdominal wall and provide early warning of post-operative infection.

Facing the challenge of ovarian cancer

A current ARC initiative focuses on the treatment of ovarian cancer. Ovarian cancer is particularly difficult to detect, which means it has a significantly

higher mortality rate compared to other cancers. This poses a big challenge for ARC.

"Ovarian cancer is usually discovered very late, stages three and four, by which time there are limited treatment options," says Hadas Heger, Project Manager, ARC Innovation. "This means there is also little data for us to work with, and almost no data from women surviving for longer than five years after diagnosis. It's a difficult area in which there have been no significant advances in treatment for the past 20 years."

Heger and her project team have specifically chosen this challenging branch of oncology not only to improve patient outcomes, but also to provide proof-of-concept for ARC's federated-learning approach. By incorporating data from other care providers, ARC's goal is to create a cohort large enough to develop AI algorithms capable of achieving a breakthrough in this challenging field.

Using Google Cloud to bridge the data gap

Just six months into its ovarian cancer program, however, the ARC team realized that no in-house solution would enable them to share patient data between medical centers internationally while keeping the privacy of their patients. With the privacy and security of patient data a primary concern, ARC turned to [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>).

"We wanted to share data and collaborate with medical institutions across the world, but it was difficult to do so while maintaining patients' privacy and data security," says Kacen. "With our federated approach, built on Google Cloud, we are able to collaborate keeping our patient data on our premise and respecting patient privacy and complying with health data regulations."

The team is currently in the process of using Google Cloud tools to develop a suite of AI tools, ranging from data dashboards to generative AI and machine learning models, which they hope will help predict the most effective treatment protocols for individual patients.

"We wanted to
share data and
collaborate with
medical
institutions
across the
world, but it was
difficult to do so
while
maintaining
patients' privacy
and data

security. With
our federated
approach, built
on Google Cloud,
we are able to
collaborate
keeping our
patient data on
our premise and
respecting
patient privacy
and complying
with health data
regulations."

—Eran Kacen, COO & CTO,
ARC

With the help of Looker Studio
(<https://cloud.google.com/looker-studio>), the team has
developed the Quick Clinical Insights dashboard,
which amalgamates patient data from six screens.
This dashboard not only generates a 'single view of
the truth,' it also visualizes disease status
parameters alongside treatment progression,
making it clear whether treatment is working and
recommends the next steps.

It is also now relatively easy to generate additional
machine learning models to support clinicians'

decision-making, thanks to [BigQuery ML](https://cloud.google.com/bigquery/docs/bqml-introduction)
(<https://cloud.google.com/bigquery/docs/bqml-introduction>)

Through data sharing with leading institutions in the US, Heger and her team now hope to create an AI data model that will enable clinicians to predict life-threatening blood clots during chemotherapy treatment and post-surgery, as well as a model to predict how individual patients are likely to react to chemotherapy.

ARC Innovation at Sheba Medical C...



**Sharing patient data to
democratize and scale research**

"The reason we chose to work with Google Cloud is because it understands the sensitivity of preserving patients' privacy and security when sharing data across jurisdictions," says Heger. "Creating an 'old fashioned' central database would be impossible, because data regulations are different in every country, but Google Cloud and its federated approach gave us a solution."

Thanks to Google Cloud, each institution within ARC's federated learning model can now store its data within its own jurisdiction and comply with local regulation, but that data can still be used to train AI models collecting the algorithm weights only from each site and not the data itself.

"The crucial element for us moving to the cloud is data security," explains Kacen. "Google Cloud is enabling us to upload raw patient records, share, and query them in real time without risking the patient's privacy and data security."

"Google Cloud is
enabling us to
upload raw
patient records,
share, and query
them in real time
without risking

the patient's privacy and data security."

—Eran Kacen, COO & CTO,
ARC, Sheba Medical
Center

Alongside Looker and BigQuery, the ARC team is also leveraging Google Cloud tools such as AutoML (<https://cloud.google.com/automl>) to create data dashboards quickly and efficiently.

"With Looker, we have been able to create data dashboards showing how patients react to treatment," says Heger. "These are being used to create precision treatment protocols for ovarian cancer patients. Practitioners can visualize the data easily and make data-based decisions about surgery, chemotherapy and other aspects of care planning. This capability changed how the department is working and now they have become a data-driven department."

With 300,000 women (<https://www.wcrf.org/cancer-trends/ovarian-cancer-statistics/>)

receiving an ovarian cancer diagnosis annually across the world, and patient data in such short supply, this is a prime example of how the Sheba team is using AI in an effort to help save lives.

"With Looker, we have been able to create data dashboards showing how patients react to treatment.

These are being used to create precision treatment protocols for ovarian cancer patients.

Practitioners can visualize the data easily and make data-based decisions about surgery, chemotherapy and other aspects of care planning."

—Hadas Heger, Project Manager, ARC

